



# DESIGN PATTERNS IN JAVA

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# **What are Design Patterns ?**

Design patterns are typical solutions to commonly occurring problems in software design. They are like pre-made blueprints that you can customize to solve a recurring design problem in your code.

# **Why Should We Have to Learn**

## **Design Patterns**

- Design patterns are a toolkit of tried and tested solutions to common problems in software design. Even if you never encounter these problems, knowing patterns is still useful because it teaches you how to solve all sorts of problems using principles of object-oriented design.
- Design patterns define a common language that you and your teammates can use to communicate more efficiently. You can say, “Oh, just use a Singleton for that,” and everyone will understand the idea behind your suggestion. No need to explain what a singleton is if you know the pattern and its name.

# **Classification Of Design Patterns**

## **1) Creational Patterns :**

provide object creation mechanisms that increase flexibility and reuse of existing code.

## **2) Structural Patterns :**

explain how to assemble objects and classes into larger structures, while keeping these structures flexible and efficient

## **3) Behavioral Patterns :**

take care of effective communication and the assignment of responsibilities between objects.